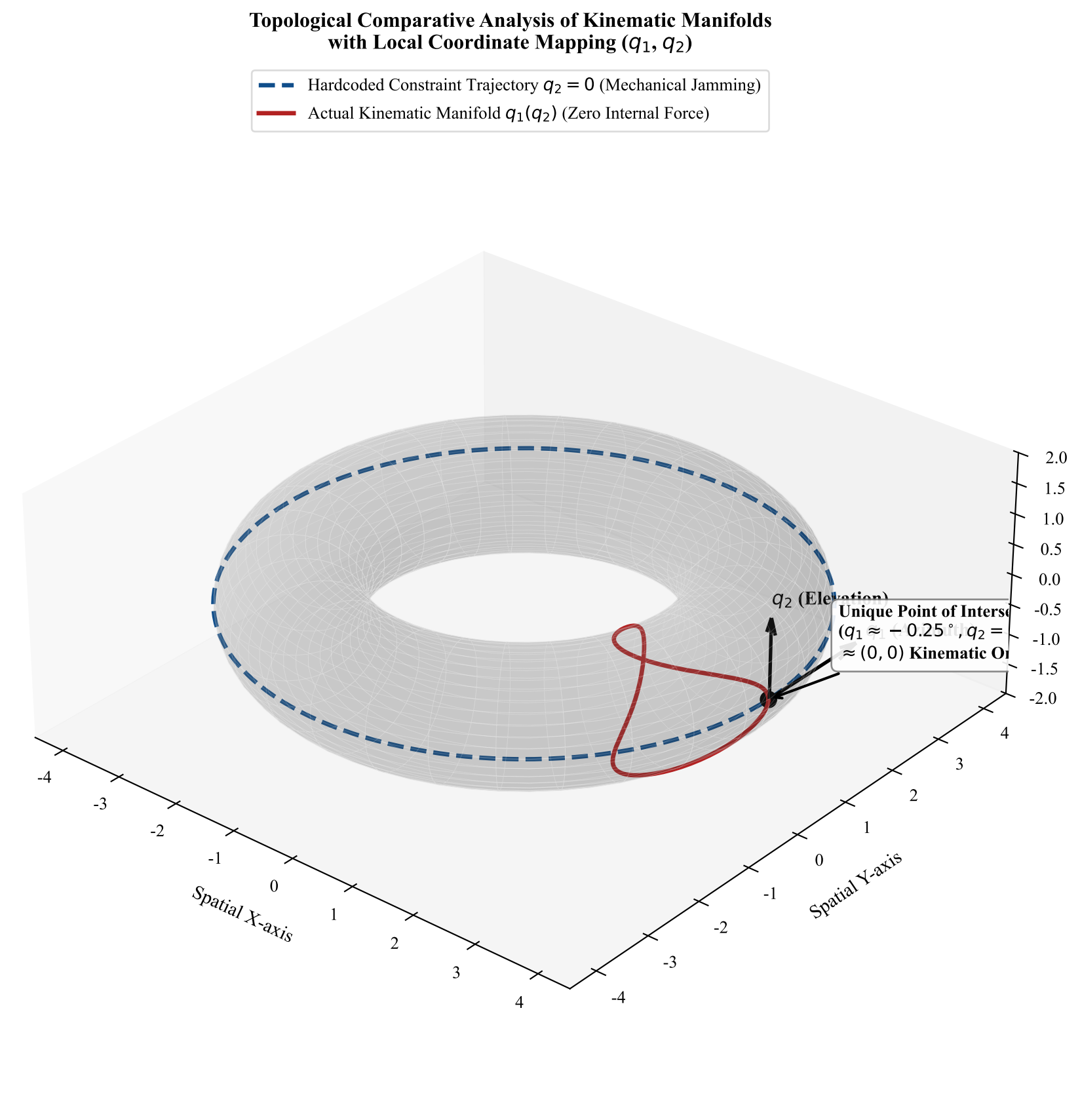

Topological Comparative Analysis of Kinematic Manifolds
with Local Coordinate Mapping ($q_1$, $q_2$)
Hardcoded Constraint Trajectory $q_2 = 0$ (Mechanical Jamming)
Actual Kinematic Manifold $q_1(q_2)$ (Zero Internal Force)
$q_2$ (Elevation)
Unique Point of Intersection
($q_1 \approx -0.25^\circ$, $q_2 \approx 0$)
$\approx (0, 0)$ Kinematic Origin
Spatial X-axis
Spatial Y-axis
-4
-3
-2
-1
0
1
2
3
4
2.0
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
-2.0